Airline Flight Operations Analytics Platform

End-to-End Data Engineering Project using Databricks and SQL

Architecture: Medallion Architecture (Bronze, Silver and Gold)

1. Project Title

Airline Flight Operations Analytics Platform using Databricks – Medallion Architecture.

2. Project Overview

This project builds an end-to-end data engineering platform to analyze airline flight operations using Databricks and SQL. It follows the Medallion Architecture: raw flight data is loaded into Bronze, cleaned and validated in Silver, and transformed into business-ready summaries in Gold. The Gold data supports a dashboard for exploring flight performance, airline operations, route performance, delays, revenue and data quality.

3. Business Problem

Raw airline data can contain missing values, duplicate records, inconsistent text formats and invalid information. Analyzing it directly can produce unreliable reports. This project creates a repeatable pipeline to preserve source data, clean and validate records, prepare summary tables and present results in a dashboard.

Project objectives include loading raw flight data, preserving source records, cleaning and validating records, separating rejected flights, creating business summary tables, checking reconciliation and visualizing results.

4. Dataset Description

The project uses the synthetic dataset **airline_flights_raw.csv**, containing **1,365 rows and 20 columns** representing domestic Indian flights during 2025. The dataset is synthetic and does not represent real airline or passenger operations.

Fields include flight_id, flight_number, airline, aircraft_id, aircraft_type, flight_date, day_of_week, origin_city, destination_city, origin_region, distance_km, scheduled_departure, actual_departure, delay_minutes, delay_reason, seat_capacity, passengers, avg_fare, ticket_revenue and fuel_cost.

The source data includes practice data-quality issues such as extra spaces, mixed capitalization, missing values, duplicate flights, passengers exceeding seat capacity and negative fares.

5. Technologies Used

Databricks; SQL; Lakeflow Spark Declarative Pipelines; Unity Catalog and Volumes; Medallion Architecture; and Databricks AI/BI Dashboards. (List Power BI only if you also used it.)

6. Medallion Architecture

Bronze preserves raw source records. Silver cleans and validates records, separating clean flights from rejected records. Gold creates business summaries for reporting.

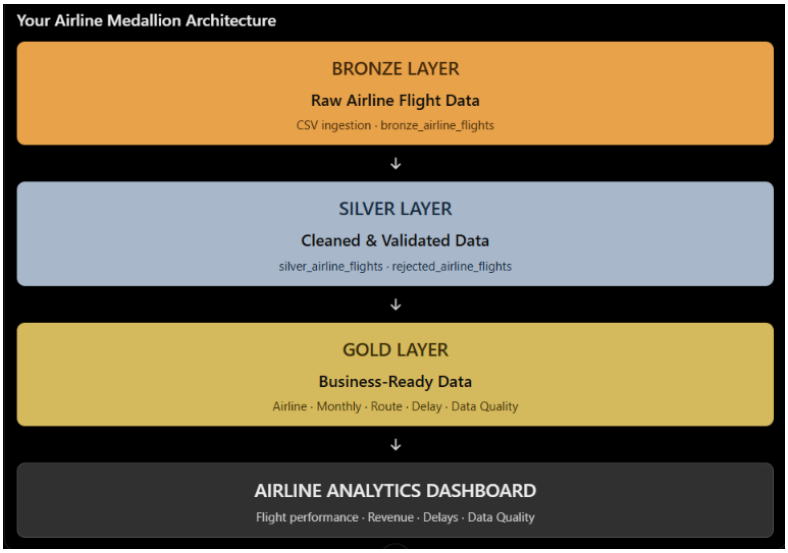


Figure 1. Medallion architecture for the Airline Flight Operations project.

7. Bronze Layer – Raw Data Ingestion

The Bronze layer is the first stage of the pipeline. The raw CSV is stored in a Unity Catalog Volume and ingested into **bronze_airline_flights**. The Bronze layer preserves source records and may include source-file information and a load timestamp. The source dataset contains 1,365 rows; the actual table count should be checked in Databricks.

8. Silver Layer – Data Cleaning

The Silver layer standardizes and validates Bronze data. The project uses **silver_airline_flights** for clean records and **rejected_airline_flights** for invalid records with a rejection reason.

Cleaning activities include deduplicating by flight_id, trimming spaces, standardizing text capitalization, filling selected missing values, calculating missing delay minutes using scheduled and actual departure times, calculating missing ticket revenue using passengers multiplied by average fare, and creating a delay category.

Delay categories: 15 minutes or less (including early departures) = On Time; 16–60 minutes = Minor Delay; more than 60 minutes = Major Delay.

Invalid records such as passengers exceeding seat capacity, negative passenger counts or negative average fares are directed to the rejected output according to the project validation rules.

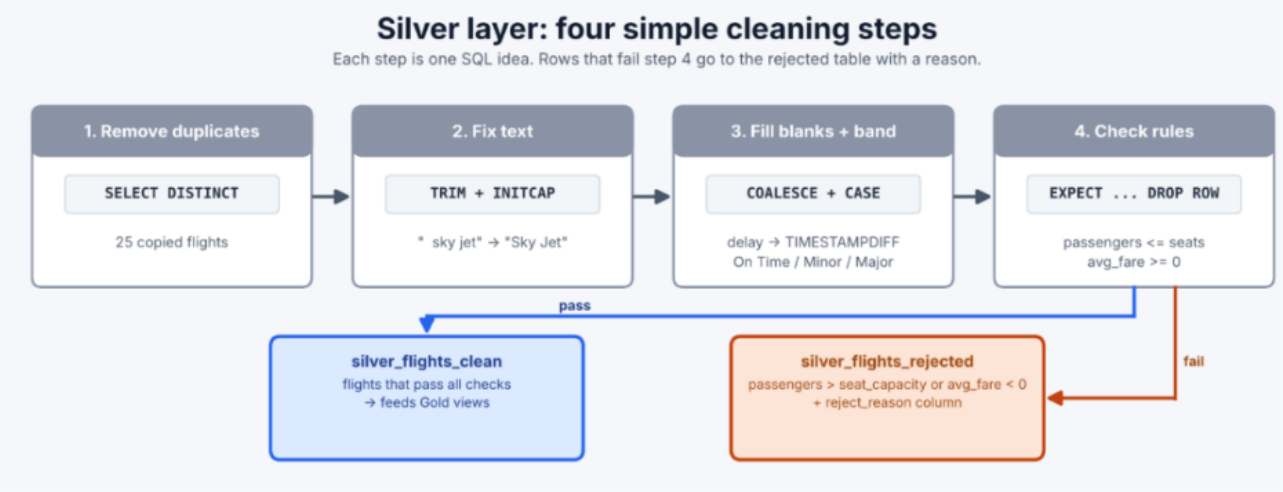


Figure 2 – The four Silver cleaning steps

Figure 2. Four Silver-layer cleaning steps and clean/rejected outputs.

9. Gold Layer – Business Analysis

The Gold layer contains business-ready summary tables created from clean Silver records.

Gold table	Purpose
gold_airline_summary	Flight, passenger, revenue, load-factor, on-time and delay metrics by airline.
gold_monthly_flights	Monthly flight, passenger, revenue and delay trends.
gold_route_summary	Flight, passenger and revenue analysis by origin and destination.
gold_delay_summary	Flight counts and average delay by delay category and reason.
gold_data_quality	Bronze, clean, rejected and duplicate counts with reconciliation checks.

Key calculations include ticket revenue = passengers × average fare; load factor = total passengers × 100 ÷ total seat capacity; and on-time percentage = flights with delay of 15 minutes or less × 100 ÷ total flights.

10. Dashboard Screenshots

The completed dashboard contains three pages. The following screenshots show the dashboard pages provided for this project.

Page 1 – Airline Flight Operations Overview

The overview page displays total flights, passengers, ticket revenue, duplicate count, Bronze rows, clean flights, rejected records and delayed flights. It also includes monthly ticket revenue, on-time percentage by airline, flights by delay category and data-quality issues.

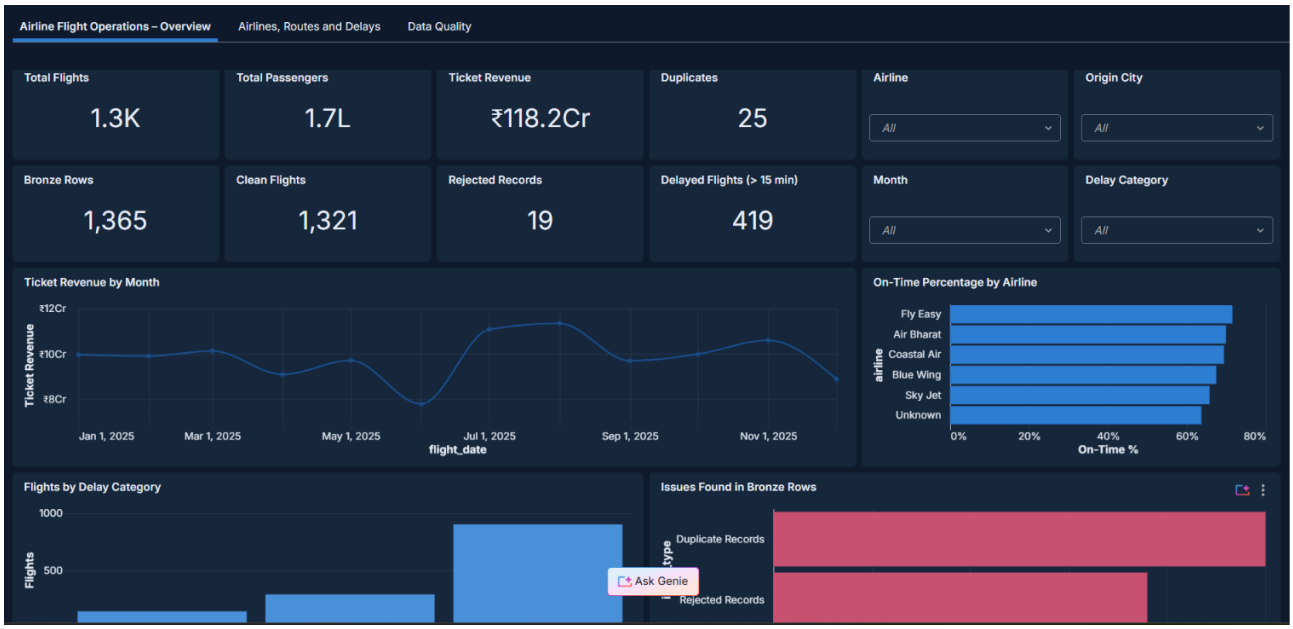


Figure 3. Airline Flight Operations – Overview dashboard.

Page 2 – Airlines, Routes and Delays

This page compares airline-wise flight performance, including flights, passengers, ticket revenue, average delay, on-time percentage, delayed flights and load factor. It also shows the top routes by revenue and the number of flights in each delay category.

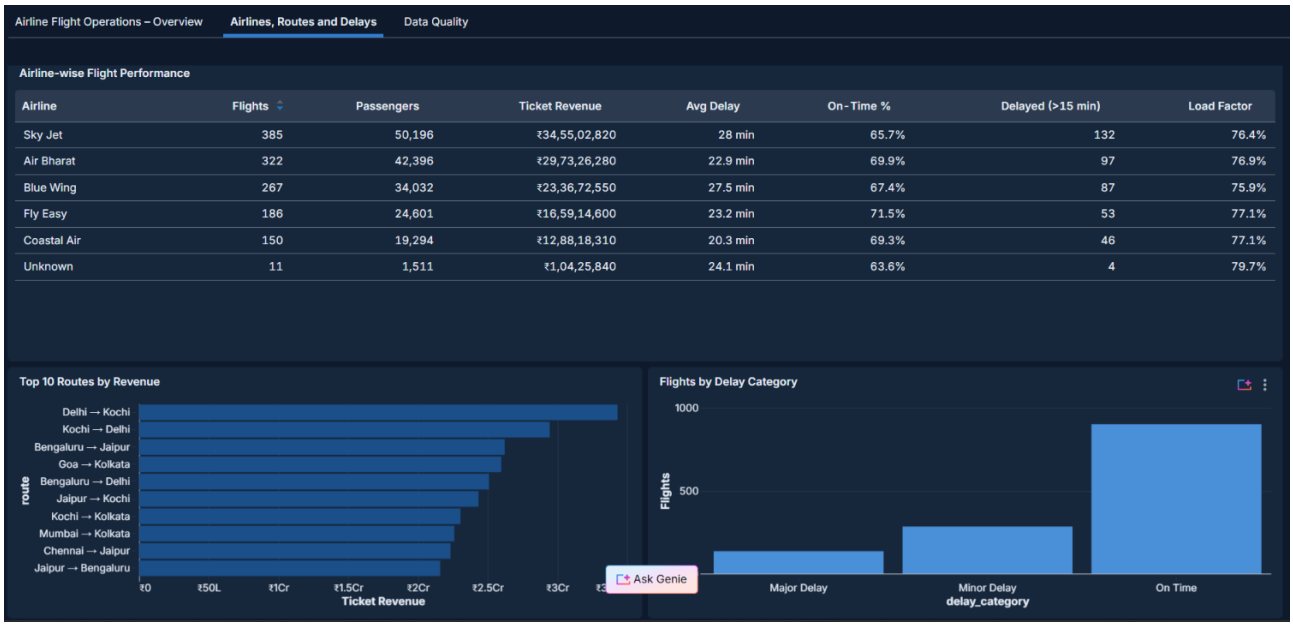


Figure 4. Airlines, Routes and Delays dashboard.

Page 3 – Data Quality

This page presents Bronze rows, clean flights, rejected records, duplicate records, reconciliation difference and reconciliation status. The screenshot shows 1,365 Bronze rows, 1,321 clean flights, 19 rejected records, 25 duplicate records and a reconciliation difference of 0 with status Passed.

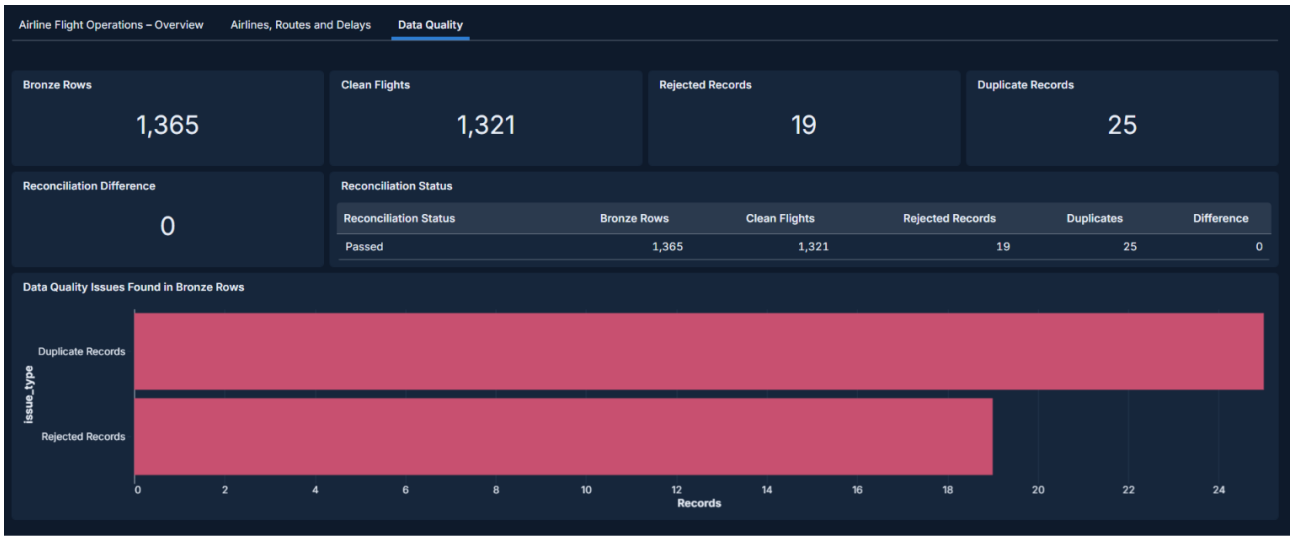


Figure 5. Data Quality dashboard.

11. Key Business Insights

Based on the supplied dashboard screenshots, the following observations can be reported:

• The dashboard shows 1,365 Bronze rows, 1,321 clean flights, 19 rejected records and 25 duplicate records removed.
• The reconciliation difference is 0 and the displayed reconciliation status is Passed.
• Sky Jet has the highest flight count in the airline-wise table, with 385 flights.
• Fly Easy has the highest on-time percentage in the airline-wise table, at 71.5%.
• Delhi → Kochi is the highest-revenue route shown in the Top 10 Routes chart.
• The overview shows 419 delayed flights (more than 15 minutes).

These findings are based on the dashboard screenshots supplied for this documentation. They describe the displayed results and should be checked again if the underlying tables or dashboard are refreshed.

12. Conclusion

This project demonstrates how raw airline flight data can be organized and transformed into business-ready information using Databricks, SQL and the Medallion Architecture. Bronze preserves source data, Silver cleans and validates records, and Gold prepares summaries for analysis. The three-page dashboard presents flight operations, airline and route performance, delays, revenue and data quality.

The project provides practical experience with SQL transformations, data pipeline development, data-quality validation, analytical table creation and dashboard reporting.

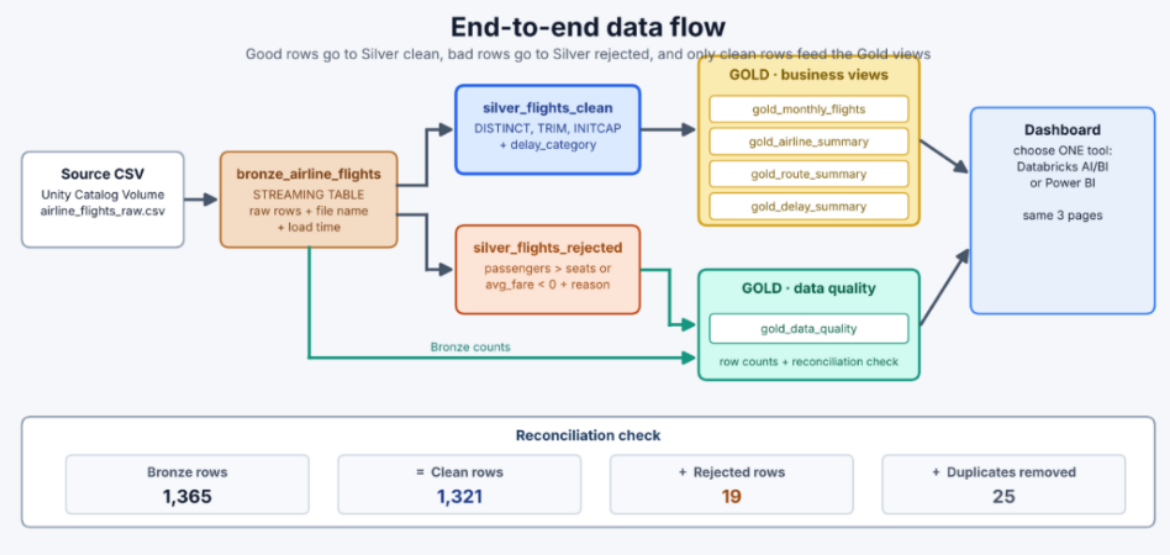


Figure 1 – End-to-end data flow and reconciliation check

Figure 6. End-to-end data flow and reconciliation check.

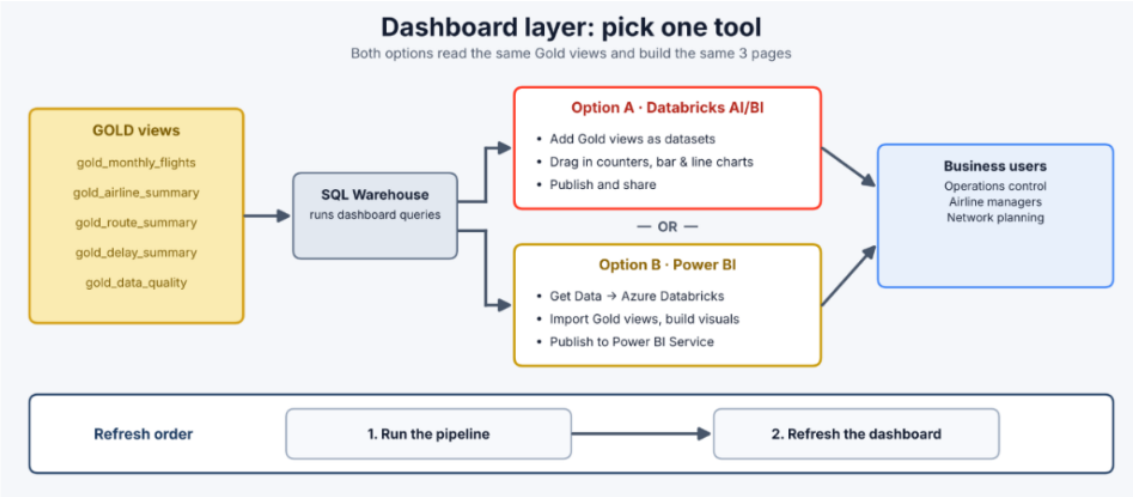


Figure 3 – Dashboard layer: one tool, same Gold tables

Figure 7. Dashboard layer design and dashboard-tool options from the project reference.

Project: Airline Flight Operations Analytics Platform
Architecture: Medallion Architecture (Bronze, Silver and Gold)
Primary tools: Databricks and SQL